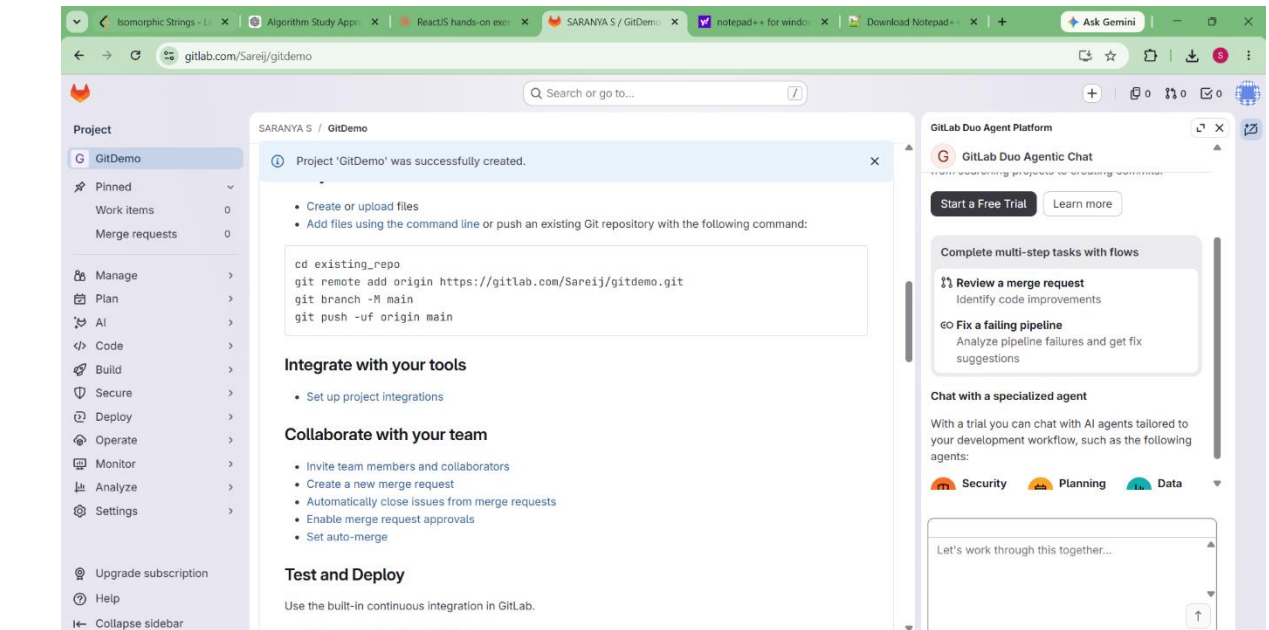


1. Git-HOL



```
MINGW64:/c/Users/saranya/GitDemo

No commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   welcome.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git status
On branch master

No commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   welcome.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git commit
[master (root-commit) c101d9b] Added welcome.txt
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ cd ~/GitDemo

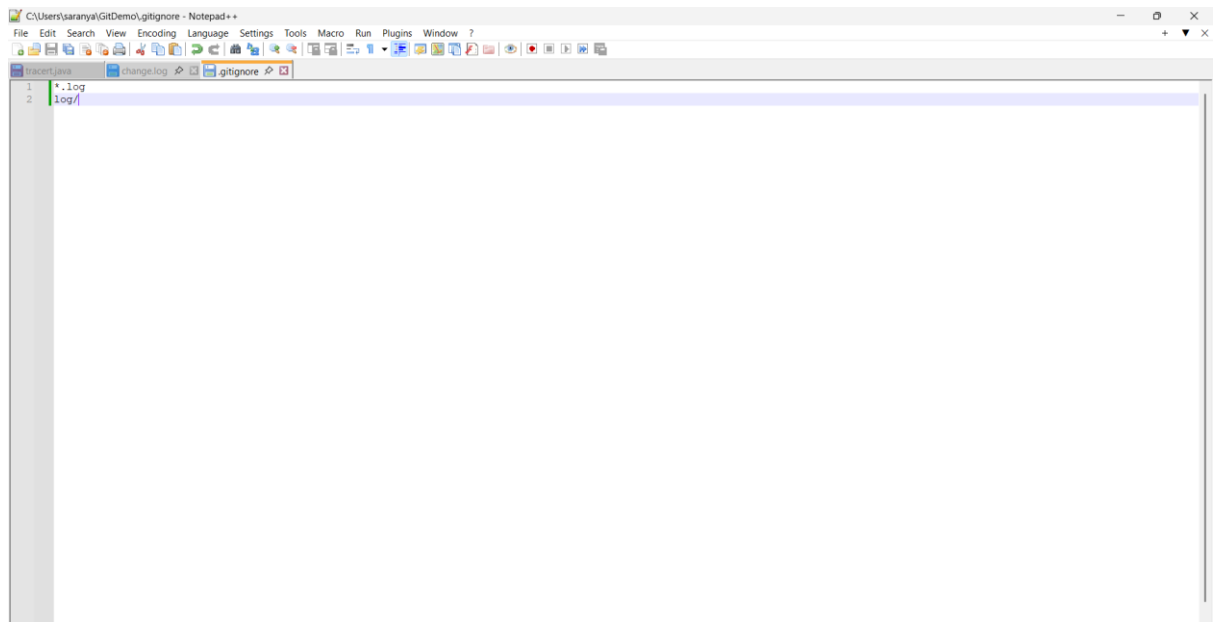
saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git remote add origin https://gitlab.com/Sareij/gitdemo.git

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git remote -v
origin https://gitlab.com/Sareij/gitdemo.git (fetch)
origin https://gitlab.com/Sareij/gitdemo.git (push)

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 2/8 bytes | 139.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/Sareij/gitdemo/-/merge_requests/new?merge_requests%5Bsource_branch%5D=master
To https://gitlab.com/Sareij/gitdemo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$
```

2. Git-HOL



```
nothing added to commit but untracked files present (use "git add" to track)
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$ notepad++ .gitignore
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked Files:
  (use "git add <file>..." to include in what will be committed)
.gitignore

nothing added to commit but untracked files present (use "git add" to track)
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$ git add .gitignore
git commit -m "Added .gitignore to ignore log files and folder"
[master c0c5b52] Added .gitignore to ignore log files and folder
1 file changed, 2 insertions(+)
create mode 100644 .gitignore
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$ git add .gitignore
git commit -m "Added .gitignore to ignore log files and folder"
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 307 bytes | 307.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote: https://gitlab.com/Sareij/gitdemo/-/merge_requests/new?merge_requests%5Bsource_branch%5D=master
remote:
To https://gitlab.com/Sareij/gitdemo.git
e101d9b..c0c5b52 master -> master
saranya@LAPTOP-K46NS3AB MINGW64 ~/GitDemo (master)
$
```

Git Exercise 3: Branching and Merging

Part A — Branching

```

c101d3b1c0c3b92 - master -> master
saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ cd ~/GitDemo
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git branch -a
* GitNewBranch
* master
  remotes/origin/master

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$ git branch
* GitNewBranch
  master

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file was created in GitNewBranch" > branchfile.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$ git add branchfile.txt
$ git commit -m "Added branchfile.txt in GitNewBranch"
[GitNewBranch 92f58ca] Added branchfile.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branchfile.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$

```

Part B — Merging

```

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
^[[200~git checkout master~
ash: '$'\E[200~git': command not found

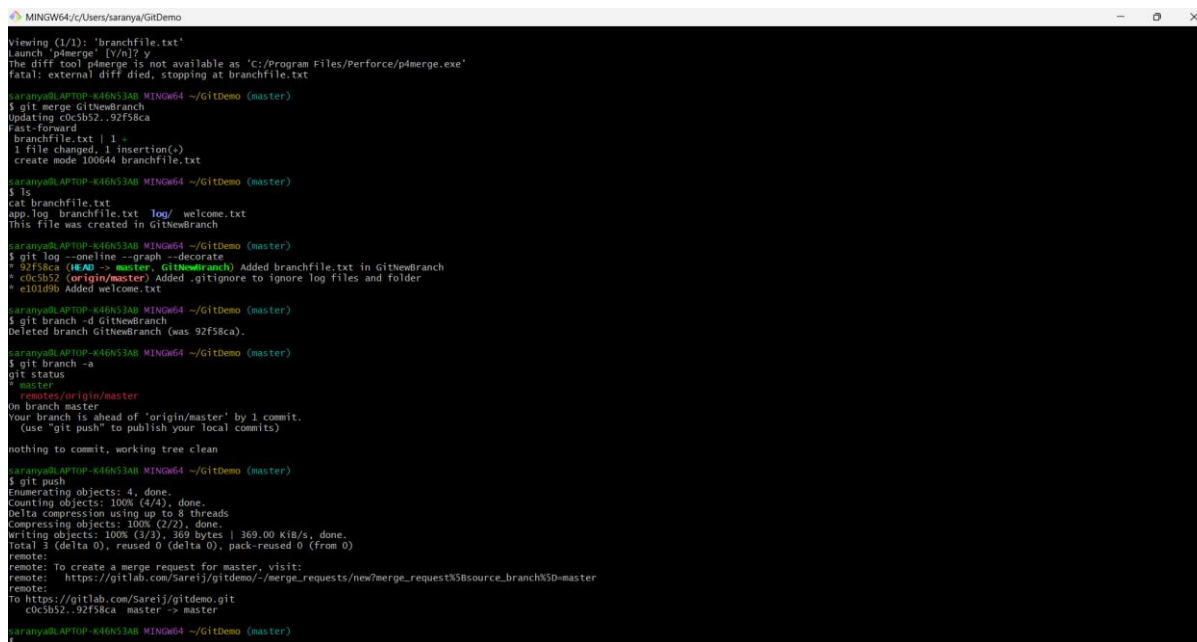
saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git branch
  GitNewBranch
* master

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branchfile.txt b/branchfile.txt
new file mode 100644
index 0000000..542dab3
--- /dev/null
+++ b/branchfile.txt
@@ -0,0 +1 @@
This file was created in GitNewBranch

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)

```



```

MINGW64/c/Users/saranya/GitDemo
Viewing (1/1): "branchfile.txt"
Launch "p4merge" [V/n]: y
The diff tool p4merge is not available as 'C:/Program Files/Perforce/p4merge.exe'
fatal: external diff died, stopping at branchfile.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating c0c3b52..92f58ca
Fast-forward
 branchfile.txt | 1 +
 1 file changed, 1 insertion(+)
create mode 100644 branchfile.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ ls
cat branchfile.txt
app.log branchfile.txt log/ welcome.txt
This file was created in GitNewBranch

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 92f58ca (HEAD -> master, GitNewBranch) Added branchfile.txt in GitNewBranch
* c0c3b52 (origin/master) Added .gitignore to ignore log files and folder
* e101d9b Added welcome.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 92f58ca).

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git branch -a
$ git status
* master
  remotes/origin/master
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 369 bytes | 369.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote: https://gitlab.com/Sareij/gitdemo/-/merge_requests/new?merge_request%5Bsource_branch%5D=master
remote:
to https://gitlab.com/Sareij/gitdemo.git
c0c3b52 master -> master

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$

```

```
MINGW64/C/Users/saranya/GitDemo
saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$ git commit -m "Added hello.xml in Gitwork branch"
[Gitwork ac07693] added hello.xml in Gitwork branch
1 file changed, 3 insertions(+)
create mode 100644 hello.xml

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$ git add hello.xml

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$ notepad++ hello.xml
git checkout master
is

notepad++ hello.xml

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$ git status
On branch Gitwork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$ git add hello.xml
git commit -m "Added hello.xml in master branch with different content"
[Gitwork b2b174f] Added hello.xml in master branch with different content
1 file changed, 3 insertions(+)
create mode 100644 hello.xml

saranya@LAPTOP-K4GN53AB MINGW64 ~/GitDemo (Gitwork)
$
```

4.ReactJS-HOL

The screenshot shows the GitLab web interface for a repository named 'gitdemo'. The interface includes a sidebar with navigation options like 'Project', 'Work items', 'Merge requests', 'Manage', 'Plan', 'AI', 'Code', 'Merge requests', 'Repository', 'Branches', 'Commits', 'Tags', 'Repository graph', 'Compare revisions', 'Snippets', 'Build', 'Secure', 'Upgrade subscription', 'Help', and 'Collapse sidebar'. The main content area displays a notification: 'You pushed to master 2 minutes ago' with a 'Create merge request' button. Below this, the 'Files' section shows a list of files: '.gitignore', 'branchfile.txt', 'hello.xml', and 'welcome.txt'. A 'Resolved merge conflict in hello.xml' notification is also present, showing the commit hash '259fc236' and a 'History' button. A table lists the files and their commit history:

Name	Last commit	Last update
.gitignore	Added .gitignore to ignore log files an...	7 hours ago
branchfile.txt	Added branchfile.txt in GitNewBranch	7 hours ago
hello.xml	Resolved merge conflict in hello.xml	6 minutes ago
welcome.txt	Added welcome.txt	8 hours ago

```

MINGW64~/Users/saranya/GitDemo
saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ notepad++ hello.xml

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git add hello.xml

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git commit -m "Added hello.xml in master branch with different content"
[master bb2ecd5] Added hello.xml in master branch with different content
1 file changed, 3 insertions(+)
create mode 100644 hello.xml

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate --all
* bb2ecd5 (HEAD -> master) Added hello.xml in master branch with different content
| * ac07693 (GitWork) Added hello.xml in Gitwork branch
|
| * 92f58ca (origin/master) Added branchfile.txt in GitNewBranch
* c0c5b52 Added .gitignore to ignore log files and folder
* e101d9b Added welcome.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git merge Gitwork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.
bash: Auto-merging: command not found
bash: syntax error near unexpected token 'add/add'
bash: Automatic: command not found
bash: fix: command not found

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:

```

```

Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.
bash: Auto-merging: command not found
bash: syntax error near unexpected token 'add/add'
bash: Automatic: command not found
bash: fix: command not found

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both added:      hello.xml

no changes added to commit (use "git add" and/or "git commit -a")

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ notepad++ hello.xml

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ git add hello.xml

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master|MERGING)
$ git commit -m "Resolved merge conflict in hello.xml"
[master 259fc23] Resolved merge conflict in hello.xml

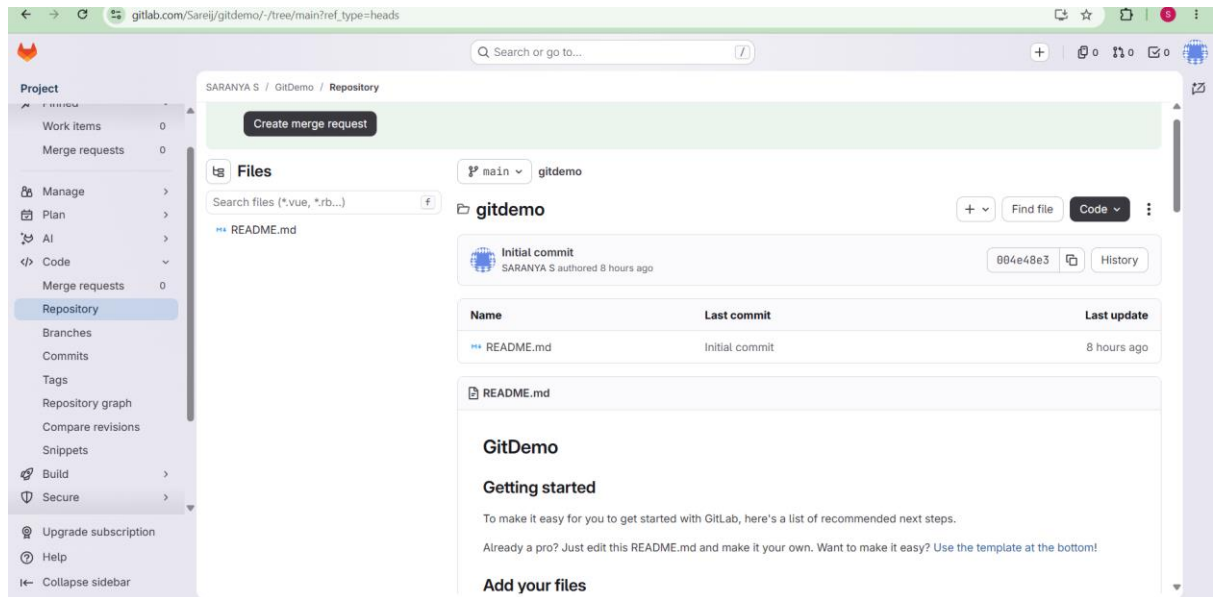
saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git status
git log --oneline --graph --decorate --all
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
* 259fc23 (HEAD -> master) Resolved merge conflict in hello.xml
|
| * ac07693 (GitWork) Added hello.xml in Gitwork branch
| * bb2ecd5 Added hello.xml in master branch with different content
|
| * 92f58ca (origin/master) Added branchfile.txt in GitNewBranch
* c0c5b52 Added .gitignore to ignore log files and folder
* e101d9b Added welcome.txt

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$

```

5. Git-HOL



```
saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git push origin master
Everything up-to-date

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ cd ~/GitDemo
git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git branch -a
* master
  remotes/origin/master

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git pull origin master
git push origin masterFrom https://gitlab.com/Sareij/gitdemo
* branch      master      -> FETCH_HEAD
Already up to date.

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ git push origin master
Everything up-to-date

saranya@LAPTOP-K46N53AB MINGW64 ~/GitDemo (master)
$ |
```